python

Languages

- Solve problems using a computer, give the computer instructions.
- Remember our diaper-changing exercise?

Talk the talk

- Speak its language
  - High-level: Python, C++, Java
  - Low-level: machine language, computers can only execute these
  - High-level languages have to be processed into low-level before the computer can run them
  - But high-level languages can run on different kinds of computers and are easier for humans to write and read, so most programs are written in high-level

Translation

- How does high-level get translated into low-level?
  - Interpreters and compilers!
  - Interpreter processes the program a little bit at a time and runs it
  - Compiler translates everything before running it

What is python ?

Python is a programming language that lets you work more quickly and integrate your systems more effectively.

(in other words, magic!)

http://www.python.org/

Python Vocab

- ***Program*** – A larger file of code that may contain one or more functions.
- ***Variable*** - names that you can assign values to, allowing you to reuse them later on.

  E.g.: `x = 1` or `msg = "Hi, I'm a message!"`
- ***Comments*** – These are notes ignored by the computer. In Python, comments start with a hash mark (#) and end at the end of the line.

  E.g.: `>>> x + y #both variables store user input`
- ***Operators*** – Mathematical symbols, like +, -, *, and /, but also ** (for exponents).

Python Vocab

- ***Keyword*** – Words with meaning/purpose in Python. E.g. “`and`”, “`print`”, and “`if`”.
- ***Function*** – A chunk of code that performs an action. Functions have names and are reusable. Kinds: built-in ones, and “user-defined” ones.
- ***Expression*** – Statements that produce values. Examples include 3 + 5, "Hello world!“.
- ***Error*** – When your program has a problem, the command area will provide an error message.

What is JES?

Jython Environment for Students allows you to program and experiment with python.

JES - Jython Environment for Students - meh

File Edit Watcher MediaTools JES Functions Window Layout Help

```
1 #hohoho
2
3
```

The top part (the white window with the tiny number 1, which tells you that that's line number 1), or the "program area", works like any text editor would, where you can type stuff and save it under some file name so you can close it and pull it back up later.

Load Program UNLOADED Watcher Stop

```
>>>
>>>
>>>
>>>
>>>
>>>
>>>
>>>
>>>
```

The bottom part (in black), the "command area", is the brains of the operation.

For help on a particular JES function, move the cursor over it Explain <click> Line Number:3 Position: 1

JES - Jython Environment for Students - meh

File Edit Watcher MediaTools JES Functions Window Layout Help

```
1 #hohoho
2
3
```

After you are done writing a program here

Click Load Program to compile the program

Load Program Watcher Stop

```
>>>
>>>
>>>
>>>
>>>
>>>
>>>
>>>
>>>
```

For help on a particular JES function, move the cursor over it Explain <click> Line Number:3 Position: 1

Writing Your Program

Header Comments:

```
# file name: circ.py
# author: Durrah Almansour
# description: a program to
calculate the area of a circle.
```

Indentation

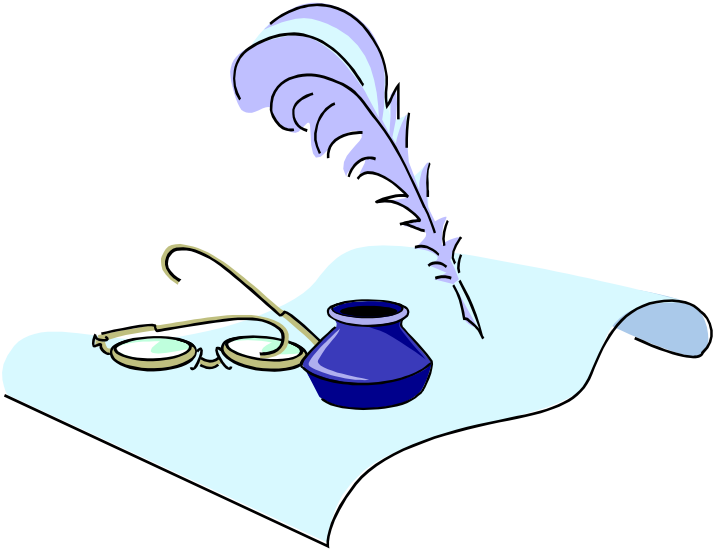
- Not an Option, but a Requirement!
- In Python, indenting specifies the “scope” of different chunks of your code.

```
def main():
  print "Hello world!"
```

The print line will be executed when you invoke main().

Writing Your Program

- Always plan what you want your program to do (pseudocode).
- Divide it into parts, it will make it easier to debug and update later.



Writing Your Program

Defining Functions:

```
def main():
    print "Hello world!"
```

- Name the function for the purpose you wrote it for.
- Don’t forget to indent the instructions that go inside the function.

Argument

A value passed to a function or method, assigned to a named local variable in the function body.

Ex.
```
def addUp(a,b):
        print("this is a+b: ", a+b)
```

#a and b are the *parameters*, 1 and 3 are the arguments we passed in.

#The function outputs : this is 1+3: 4

http://docs.python.org/glossary.html#glossary